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Essentials of Data Science Laboratory - 2304102L - 2026

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5.1. Practice Lab Assignment

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5.2.2. Histogram of passenger information ...

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5.1.1. Stacked Plot

08:44

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

Create a stacked area plot using the data.

Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.

Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

Submit

1import matplotlib.pyplot as plt

2import pandas as pd

3

4# Data for Months and Temperature for three cities

5data = {

6Month: ['January', 'February', 'March',

April, 'May', 'June', 'July', 'August',

September, 'October', 'November', 'December'],

7City_A_Temperature: [5, 7, 10, 13, 17, 20,

Average time0.832 s832.00 ms

Maximum time0.832 s832.00 ms

1 out of 1 shown test case(s) p

Test case 1832 msDebug

Expected output

Actual output

Temperature Variation

Temperature Variation

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5.2.1. Titanic Dataset

01:30

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (0-1000).

titanicDat...

Submit

1

2

3

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8

9

Average time

3.139 s

3139.00 ms

Maximum time

3.139 s

3139.00 ms

1 out of 1 shown test case(s) p

Test case 1

3139 ms

Debug

Expected output

Actual output

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5.2.2. Histogram of passenger information of Titanic

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**.
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below.

[illegible]

Sample Data:

 Histogra...

Submit

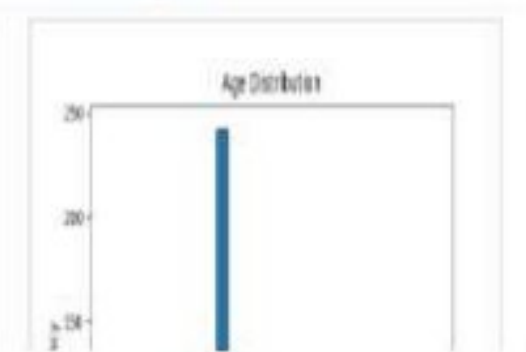
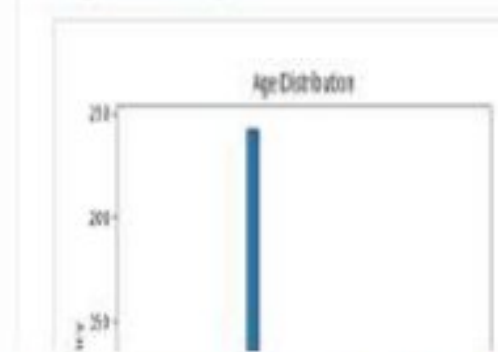
```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(),
9 inplace=True)
```

Average time **0.897 s**  Maximum time **0.897 s**   1 out of 1 shown test case(s) passed

✔ Test case 1 897 ms

Expected output

Actual output



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5.2.3. Bar plot of survival rate of passengers01:34

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to 'bar'.
3. Add the title "Survival Count" to the chart.
4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
a	u	l	a	e	g	i	a	i	a	a	m
s	r	c	m	x	e	b	r	k	r	b	b
s	v	s	e		S	c	e	e	e		k
e	e				p	h	t				e
r	d										d

BarPlotOf...Submit

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(),
9                    inplace=True)
```

Average time0.898 s898.00 msMaximum time0.898 s898.00 ms

1 out of 1 shown test case(s) passed

Test case 1898 msDebug

Expected output

Actual output

Terminal

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5.2.5. Bar Plot for Survival by Pclass

01:24

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts().

2. Use a stacked bar chart to display the survival counts.

3. Add the title "Survival by Pclass" to the chart.

4. Label the x-axis as 'Pclass' and the y-axis as 'Count'.

5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	P	T	F	C	E
a	u	cl	a	e	g	i	a	a	i	a	a	m
s	r	a	m	x	e	b	r	r	c	r	b	b
e	v	s	e				c	k	e	e	i	a
n	e						h	t			n	r
g	d											e
e												d
r												
d												

BarPlotOf...

Submit

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# Data Cleaning

8data['Age'].fillna(data['Age'].median(),

inplace=True)

Average time0.942 s942.00 ms

Maximum time0.942 s942.00 ms

1 out of 1 shown test case(s) p

Test case 1942 msDebug

Expected output

Actual output

Survival by Pclass

Survival by Pclass

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1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

[illegible]

Sample Data:

Submit

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(),
9 inplace=True)
```

Average time: **0.970 s** Maximum time: **0.970 s** 1 out of 1 shown test case(s) passed

Test case 1 970 ms Debug

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5.2.11. Scatter Plot for Age vs. Fare by Sur...

5.2.8. Box Plot for Age by Survived

00:48

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:
1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
a	u	l	a	e	i	a	i	i	a	a	m
s	r	s	m	x	g	b	r	k	r	b	b
e	v										
n	e										
g	d										
e											
r											
i											
d											

BoxPlotF...

Submit

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(),
inplace=True)

Average time0.852 s882.00 msMaximum time0.852 s882.00 ms1 out of 1 shown test case(s) passed

Test case 1852 msDebug

Expected outputActual output

Age by SurvivalAge by Survival

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5.2.9. Box Plot for Fare by Pclass

01:26

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:
1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

P	a	s	s	e	n	g	e	r	i	d
S	u	r	v	i	v	e	d			
P	c	l	a	s	s					
N	a	m	e							
S	e	x								
A	g	e								
S	i	b	s	p						
P	a	r	c	h						
T	i	c	k	e	t					
F	a	r	e							
C	a	b	i	n						
E	m	b	a	r	k	e	d			

Sample Data:

BoxPlotF...

Submit

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(),
inplace=True)

Average time0.786 s
Maximum time0.786 s
1 out of 1 shown test case(s) passed

Test case 1786 msDebug

Expected output

Actual output

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01:25 🔊 🌙 📝 🔗 —

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

[illegible]

Sample Data:

Submit

Debugging

3

 Debug

Actual output



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5.2.10. Scatter Plot for Age vs. Fare

5.2.11. Scatter Plot for Age vs. Fare by Sur...

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to "Age vs. Fare".
3. Label the x-axis as 'Age' and the y-axis as 'Fare'.

[illegible]

Sample Data:

Submit

Average time: **0.736 s**  Maximum time: **0.736 s**   1 out of 1 shown test case(s) passed

Test case 1 738 ms Debug

Expected output



Actual output



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5.2.9. Box Plot for Fare by Pclass

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5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:
1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.
The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
a	u	l	a	e	g	i	a	i	a	a	m
s	r	c	m	x	e	b	r	c	r	b	b
e	v	s	e				c	k	e		
n	e						h	e			
g											
e											
r											
d											

AgeFareS...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(),
inplace=True)

Average time0.815 s
Maximum time0.815 s
1 out of 1 shown test case(s) p

Test case 1 815 ms
Expected output
Actual output

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